strating state-of-the-art performance on the MLEBench benchmark (Chan et al., 2025), designed for machine learning engineering tasks. In AIDE, each node represents a potential solution state with a corresponding scalar evaluation score (e.g., validation accuracy). Nodes are iteratively selected for further debugging or refinement based on these scores. Inspired by this approach, we integrate a similar tree search-based exploration strategy within our automated scientific discovery framework, adapting it specifically to the multi-stage nature of scientific experimentation, as detailed in §3.

3. The AI Scientist-v2

We now describe the major innovations introduced in The AI Scientist-v2 relative to The AI Scientist-v1 (Lu et al., 2024). The most significant improvement is the move towards greater autonomy and generalization, starting a more general idea generation phase (§3.1) and eliminating the reliance on fixed, human-authored template code for experimentation. This process begins with generalized idea generation, producing an initial concept, which then feeds into the experimentation phase (§3.2). To manage this, we introduce two critical features in the experimentation phase: *coarse-grained experiment management* and *agentic tree search-based exploration*. Additionally, we integrate Vision Language Models (VLMs) into the experimental and review phases (§3.4). Finally, we streamline the manuscript writing phase by replacing the incremental, Aider-based (Gauthier, 2024) iterative writing approach of The AI Scientist-v1 with a simpler, single-pass generation followed by a separate reflection stage powered by reasoning models such as o1 (OpenAI, 2024). We include a full list of sampling hyperparameters and models used in Appendix A and the prompts used for The AI Scientist-v2 in Appendix B.

3.1. More General Idea Generation

A key conceptual shift in The AI Scientist-v2 is the approach to research idea generation. Unlike the predecessor system, which primarily focused on proposing incremental modifications or extensions based on an existing codebase, The AI Scientist-v2 adopts a process that begins at a higher level of abstraction. The system is prompted to engage in more open-ended thinking about potential research directions, hypotheses, and experimental designs, akin to formulating a research abstract or grant proposal before committing to a specific implementation.

This approach encourages the exploration of potentially more novel or foundational ideas, rather than being constrained by the structure and topics of pre-existing code. It aligns more closely with how researchers often develop broader research visions, starting with abstract concepts and assessing novelty and feasibility before diving into specific implementations. Crucially, this generalized idea generation phase integrates literature review tools, such as Semantic Scholar, in the loop. The system can query the literature database during the idea formulation process to assess the novelty of a proposed concept and identify relevant prior work. This allows for more informed decisions about pursuing a particular research avenue, ensuring ideas are grounded in the existing scientific landscape from the outset, rather than relying solely on post-hoc checks.

3.2. Removing Template Dependency

Following the improved idea generation phase, The AI Scientist-v2 proceeds with experimentation. Beyond the code-conditioned idea generation, The AI Scientist-v1 also depended on the predefined template code as a starting baseline implementation. The LLM-driven code changes were then limited to sequential code adaptations. We now outline our strategy for eliminating this limitation, thus improving the system's flexibility and autonomy.

3.2.1. Experiment Progress Manager

Real-world scientific experimentation typically proceeds through distinct stages, from initial feasibility assessments to detailed ablation analyses. To emulate this structured approach, we introduce an